

Consider the curve defined implicitly by the equation

$$xy^3 + 2xy = 6$$

1. Use implicit differentiation to find an expression for $\frac{dy}{dx}$ that depends on x and y . Show all of your work and thinking clearly, using correct notation.
2. Notice that the point $(2, 1)$ lies on the curve $xy^3 + 2xy = 6$. Find the exact slope of the tangent line to curve at $(2, 1)$. Show clearly how you determined this value, and label your work with proper notation.